Nathan S. Martin

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EDUCATION

Bachelor of Software Engineering | University of Waterloo *Sep 2023 - Present*

- Joint Honours in Combinatorics and Optimization | Specialization in Artificial Intelligence
- **97.4%** Major Average | **95.0%** Cumulative Average | **4.0** GPA

SKILLS

Languages: C++, C, Python, C#, Java, Scala, Kotlin, Shell Script, VHDL, SQL, JavaScript / TypeScript

Technologies: PyTorch, OpenCV, AForge.NET, Git, SVN, MongoDB, Firebase, Docker, REST APIs, Unity, Unreal Engine, Pygame, Panda3D, LaTeX, Fusion360, Raspberry Pi, Arduino, ESP32, Nvidia Jetson, KiCad, I2C, SPI, FreeRTOS

EXPERIENCE

Ford Motor Company | Automotive Software Developer *Jan 2025 - Apr 2025*

- Designed and Implemented C++ hardware abstraction layers (HALs) and Java middleware to interface with vehicle ECUs
- Built Android applications in Kotlin for the infotainment system, enhancing driver-assist features
- Developed production-grade software, deployed on millions of vehicles, ensuring reliability and real-world performance

University of Waterloo | Formal Methods Research Assistant *Jan 2025 – Present*

- Worked with Dr. Nancy Day on a formal verification tool for validating first-order logic and program correctness proofs
- Optimized speed by over 10x through designing efficient algorithms and implementing multithreaded test execution
- Collaborated with researchers and graduate students, presenting findings to refine the tools usability and speed

Connect Tech Inc | Embedded Software Developer *May 2024 - Aug 2024*

- Contributed to the MLCommons **MLPerf Inference** Working Group, working on AI benchmarking for edge computing
- Developed BSPs for Nvidia Jetson carrier boards by writing kernel drivers, device trees, and shell scripting automation
- Implemented disk encryption and OTA updates, enhancing security and serviceability of devices deployed in the field

McMaster University | Computational Physics Developer *Feb 2023 - Jun 2023*

- Co-created **DATASCI 3VG4**, an upper-year university course on developing physics engines, with Dr. James Wadsley
- Developed force and impulse-based, real time, rigid body physics simulations using Panda3D and Python
- Designed and implemented algorithms to triangulate, render, and detect collisions of prisms and ellipsoids

PROJECTS

Skatelligence | *AI-Powered Figure Skating Analysis* **Website**

- Created an LSTM-based recurrent neural network (RNN) with PyTorch to classify figure skating jumps with 80% accuracy
- Built a generative adversarial network (GAN) to generate synthetic IMU data, improving identification accuracy by 50%
- Programmed an ESP32 microcontroller to use a RTOS for multithreaded IMU data collection and transmission to a server

Waterloo Aerial Robotics Group

- Leveraged PyTorch and YOLOv8 classification model to extract landing pad locations from real-time video
- Engineered the autopilot decision-making algorithm, integrating telemetry, computer vision data, and mission commands
- Interfaced with the ArduPilot flight controller using MAVLink to monitor the drone's status and transmit commands

Rubik's Cube Solving Robot **Website**

- Designed and built a Rubik's Cube solving robot with a custom algorithm, 25% more efficient than speedcubers.
- Developed a reliable and fast image recognition algorithm to scan the Rubik's Cube, powered by AForge.NET

Refashion | *Wardrobe Organizing Smart Mirror*

- Used OpenCV and a YOLOv8 image segmentation model for clothing detection and identification.
- Integrated LLM-powered personalized outfit recommendations into a Raspberry Pi powered smart mirror